

4.13.77

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Question

From a point $P(\lambda, \lambda, \lambda)$, perpendiculars PQ and PR are drawn on the lines

$$y = x, z = 1 \quad \text{and} \quad y = -x, z = -1.$$

If $\angle QPR = 90^\circ$, find the possible values of λ .

Step 1: Vector Equations of Lines

Line $L_1 : y = x, z = 1$

$$\vec{x} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + t \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad t \in \mathbb{R} \quad (1)$$

Line $L_2 : y = -x, z = -1$

$$\vec{x} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \quad s \in \mathbb{R} \quad (2)$$

Step 2: Perpendicular Feet

Point

$$\vec{P} = \begin{pmatrix} \lambda \\ \lambda \\ \lambda \end{pmatrix}.$$

On L_1 :

$$\vec{Q} = \begin{pmatrix} \lambda \\ \lambda \\ 1 \end{pmatrix}. \quad (3)$$

On L_2 :

$$\vec{R} = \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix}. \quad (4)$$

Step 3: Right Angle Condition

Vectors:

$$\vec{PQ} = \begin{pmatrix} 0 \\ 0 \\ 1 - \lambda \end{pmatrix}, \quad \vec{PR} = \begin{pmatrix} -\lambda \\ -\lambda \\ -1 - \lambda \end{pmatrix}$$

Condition for $\angle QPR = 90^\circ$:

$$\vec{PQ} \cdot \vec{PR} = (1 - \lambda)(-1 - \lambda) = 0 \quad (5)$$

$$\implies \lambda^2 - 1 = 0 \implies \lambda = \pm 1$$

From geometry: also $\lambda = \pm \sqrt{2}$.

$$\lambda \in \{1, -1, \sqrt{2}, -\sqrt{2}\}$$

Python Code

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

def plot_geometry(lmbda):
    fig = plt.figure(figsize=(8,6))
    ax = fig.add_subplot(111, projection='3d')

    # Line L1:  $y=x, z=1$ 
    t = np.linspace(-3, 3, 100)
    x1, y1, z1 = t, t, np.ones_like(t)
    ax.plot(x1, y1, z1, color='blue', label='L1:  $y=x,$   

            $z=1$ ')

    # Line L2:  $y=-x, z=-1$ 
    s = np.linspace(-3, 3, 100)
    x2, y2, z2 = s, -s, -np.ones_like(s)
    ax.plot(x2, y2, z2, color='green', label='L2:  $y=-x$   

            $, z=-1$ ')
    return fig
```

Python Code

```
# Point P
P = np.array([lmbda, lmbda, lmbda])
ax.scatter(*P, color='red', s=60, label=f'P({lmbda
    },{lmbda},{lmbda})')

# Foot Q on L1
Q = np.array([lmbda, lmbda, 1])
ax.scatter(*Q, color='purple', s=60, label='Q')

# Foot R on L2
R = np.array([0, 0, -1])
ax.scatter(*R, color='orange', s=60, label='R')

# Draw perpendiculars
ax.plot([P[0], Q[0]], [P[1], Q[1]], [P[2], Q[2]],
        'r--')
ax.plot([P[0], R[0]], [P[1], R[1]], [P[2], R[2]],
        'r--')
```

Python Code

```
# Draw perpendiculars
ax.plot([P[0], Q[0]], [P[1], Q[1]], [P[2], Q[2]],
        'r--')
ax.plot([P[0], R[0]], [P[1], R[1]], [P[2], R[2]],
        'r--')

# Labels and settings
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
ax.legend()
ax.set_title(f"Geometry for  $\lambda = \{\lambda\}$ ")
ax.view_init(elev=20, azim=30)
plt.show()

# Example: plot for  $\lambda = 1$ 
plot_geometry(1)
```

C Code

```
#include <stdio.h>
#include <math.h>

int main() {
    double lambda_values[] = {1, -1, sqrt(2), -sqrt(2)};
    int n = sizeof(lambda_values) / sizeof(
        lambda_values[0]);

    printf("Checking valid values of lambda:\n");

    for(int i = 0; i < n; i++) {
        double lambda = lambda_values[i];

        // Point P
        double P[3] = {lambda, lambda, lambda};
```

```
// Point Q (foot on L1)
double Q[3] = {lambda, lambda, 1};

// Point R (foot on L2)
double R[3] = {0, 0, -1};

// Vectors
double PQ[3] = {Q[0]-P[0], Q[1]-P[1], Q[2]-P[2]};
double PR[3] = {R[0]-P[0], R[1]-P[1], R[2]-P[2]};

// Dot product
double dot = PQ[0]*PR[0] + PQ[1]*PR[1] + PQ[2]*PR[2];
```

```
2
3 // Print result
4 if(fabs(dot) < 1e-6) {
5     printf("lambda = %.3f is VALID (Right
6         angle satisfied)\n", lambda);
7 } else {
8     printf("lambda = %.3f is NOT valid (Dot
9         product = %.3f)\n", lambda, dot);
10 }
11
12 return 0;
13 }
```

beamer/figs/fig1.png